ELECTRONICS ASSIGNMENT

AIM: To Design a system to measure the weight of food being dispensed into a pan using an appropriate load cell, ADC, connectors and a MCU.

LOAD CELL SELECTION:

There are 5 types of load cells, among which **CAPACITIVE** and **STRAIN GAUGE** are the only two that provides us with high accuracy and sensitivity, and for this application of food being dispensed, accuracy and sensitivity play a huge role.

Strain gauge load cells are the most popular ones as they are the ones most used in weighing systems and food packaging industry. So, Strain Gauge or specifically the **single point strain gauge load cell** is the best for this application as this load cell can measure the weight accurately even if the load is off the centre giving us accurate values. For the load capacity we must choose a capacity slightly higher than the added weight of food dispensed and the pan, to give us accurate readings without causing any problems to the load cell,5Kg is the best for weighing so that's the load capacity. Single point strain gauge is also known to have good resistance to mechanical disturbances due to their rigid structure.

ADC SELECTION:

As for the ADC, the best for this type of application would be the **delta - sigma ADC** which has high resolution and is highly accurate. It basically oversamples the analog data to provide accurate measurements and helps in noise reduction. **HX711** is the component that can be selected, as the signals coming out of a load cell is very poor so HX711 amplifies it and can-do digital filtering as well. It is a type of delta – sigma adc and so it has added benefits. HX711 also has 24-bit resolution which gives more accurate readings of the weight changes.

CONNECTORS AND CABLES SELECTION:

The load cell can be interfaced to the ADC using the 4-pin connector that's there in the pcb i.e. E+, E-, S+, S-. **Shielded 4-core cables** can be used to connect as they minimise **EMI** and **RFI**. Double shielded cables can also be used but its more expensive, less flexible and is generally used for long distances.

Screw terminals can be used as connectors as it is very easy to use and there is no need of any special tools and it provides us with a secure connection to the pcb. This makes the input side.

The signal coming out of HX711 is in digital form and so is less susceptible to noise and so if the MCU is integrated into the pcb then normal **pcb traces** would be proper or if its an external MCU then **jumper wires** can be used.

Decoupling capacitors can be added to reduce the noise and voltage spikes wherever needed.